

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		SLOT B
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	CSE AI	Date:	19-08-2026

Code of Conduct

I P Mohan Babu / 192472254 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P Mohan Babu

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform **Reference Resolution** by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

2. Develop an algorithm to analyze **Text Coherence** and **Discourse Structure** by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

3. Design an algorithm for a **Conversational Agent** that identifies **Dialogue Acts** from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"

Agent: "Sure, where would you like to travel?"

User: "I want to go to Chennai."

Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogue-act sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

4. Design an algorithm for **Language Generation** that converts a semantic representation into a grammatically correct sentence through **Surface Realization**. Consider the semantic input:

Action: Buy
Agent: Student
Object: Book
Tense: Past

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

5. Design an algorithm that combines **Interlingua-based Machine Translation** and **Statistical Translation Approaches**. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. Analyze the source sentence.
2. Create an Interlingua representation.

3. Generate candidate target-language translations.
4. Apply statistical scoring to select the most appropriate translation.
5. Produce the final translated sentence.

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)

Q1. Reference Resolution Algorithm

Problem

Input: A discourse containing entities and pronouns.

Output: Pronouns linked to their correct antecedents and a resolved discourse.

Pseudocode

ALGORITHM ReferenceResolution(Text)

INPUT: Text containing sentences

OUTPUT: Resolved text with pronouns replaced

```
1. sentences ← SplitIntoSentences(Text)
2. entities ← ExtractNamedEntities(sentences)
3. discourse_entities ← empty list

4. FOR each sentence IN sentences:
    mentions ← ExtractMentions(sentence)

    FOR each mention IN mentions:
        IF mention is a pronoun THEN
            candidates ← FindPreviousEntities(discourse_entities)

            FOR each candidate IN candidates:
                score ← 0

                IF Gender(mention) = Gender(candidate):
                    score ← score + 3

                IF Number(mention) = Number(candidate):
                    score ← score + 2

                IF SemanticCompatible(mention, candidate):
                    score ← score + 3

                IF Recent(candidate):
                    score ← score + 1

            antecedent ← CandidateWithHighestScore(candidates)
            Link(mention, antecedent)

        ELSE
            Add(mention, discourse_entities)

5. Replace each pronoun with its selected antecedent
6. RETURN resolved text
END ALGORITHM
```

Application

Original:

“Ravi met Arun at the library. He borrowed a book and later returned it.”

For **He**, the possible antecedents are Ravi and Arun. Both are masculine and singular, so agreement alone cannot distinguish them. The discourse context and recency preference make **Arun** a possible recent candidate, but the sentence is inherently ambiguous without additional context.

For **it**, the antecedent is clearly **book**, because a book is a suitable object of “returned.”

A reasonable resolution, assuming the intended context is that Ravi borrowed the book, is:

“Ravi met Arun at the library. Ravi borrowed a book and later returned the book.”

Validation

The resolution of **it** → **book** is highly reliable because of semantic compatibility and number agreement.

The resolution of **He** is **not uniquely determined by the supplied text**: both Ravi and Arun satisfy masculine singular agreement. Therefore, a robust reference-resolution algorithm should report ambiguity rather than falsely claiming certainty.

For example:

He → {Ravi, Arun}

it → {book}

If additional context states that Ravi borrowed the book, then:

He → Ravi

it → book

Q2. Text Coherence and Discourse Structure

Problem

Input: A sequence of sentences.

Output: Logical discourse relations and a coherent discourse structure.

Pseudocode

ALGORITHM CoherenceAnalysis(Text)

INPUT: Text containing sentences

OUTPUT: Discourse relations and discourse tree

1. sentences ← SplitIntoSentences(Text)
2. FOR each pair of consecutive sentences (S1, S2):

relation ← NONE

IF S2 contains causal markers

such as "therefore", "because", "so":

relation ← CAUSE_EFFECT

ELSE IF S2 describes an event occurring after S1:

relation ← SEQUENCE

ELSE IF S2 provides additional information about S1:

relation ← ELABORATION

StoreRelation(S1, S2, relation)

3. ConstructDiscourseTree(sentences, relations)

4. RETURN relations, discourse tree

END ALGORITHM

Application

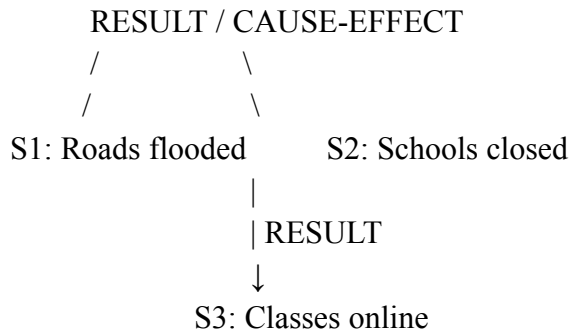
Given:

1. **S1**: “The roads were flooded after heavy rainfall.”
2. **S2**: “Therefore, schools were closed for the day.”
3. **S3**: “Students attended classes online.”

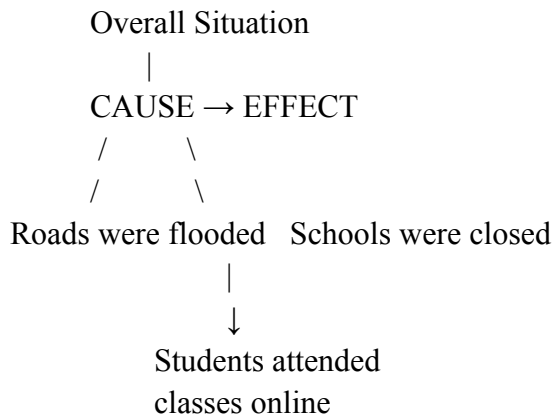
Discourse relations

- **S1 → S2**: Cause–Effect
Heavy rainfall caused flooding, and the resulting situation led to schools being closed.
- **S2 → S3**: Cause–Effect / Result
Since schools were closed, students attended classes online.

Discourse structure



A more hierarchical representation is:



Coherence justification

The discourse is coherent because each sentence logically follows from the previous situation:

Heavy rainfall → flooded roads → schools closed → online classes

The connector “**Therefore**” explicitly signals the cause–effect relationship between S1 and S2. S3 provides the consequence of the school closure. The shared topic of rainfall, schools and students creates strong discourse continuity.

Q3. Dialogue-Act Recognition Algorithm

Problem

Input: User-agent conversation.

Output: Dialogue-act label for each utterance and the resulting act sequence.

Pseudocode

ALGORITHM IdentifyDialogueAct(Utterance)

INPUT: One conversational utterance

OUTPUT: Dialogue-act label

1. IF utterance contains a request pattern
such as "Can you", "Could you", "Please":
 RETURN Request
 2. ELSE IF utterance is an information-seeking question:
 RETURN Question
 3. ELSE IF utterance provides factual information:
 RETURN Inform
 4. ELSE IF utterance confirms a completed action:
 RETURN Confirmation
 5. ELSE IF utterance describes an action performed by the agent:
 RETURN Action
 6. ELSE:
 RETURN Unknown
- END ALGORITHM

Conversation analysis

Speaker Utterance

Dialogue Act

User	"Can you book a train ticket for me?"	Request
Agent	"Sure, where would you like to travel?"	Question
User	"I want to go to Chennai."	Inform
Agent	"Your ticket has been booked."	Confirmation

Dialogue-act sequence

Request → Question → Inform → Confirmation

How the sequence helps the agent

1. **Request:** Identifies the user's goal: booking a train ticket.
2. **Question:** The agent identifies missing information needed to complete the task.
3. **Inform:** The user supplies the destination.
4. **Confirmation:** The agent communicates successful completion.

Thus, dialogue acts allow the conversational agent to track the user's intention, identify missing information, perform the requested task and confirm its completion.

Q4. Natural Language Generation and Surface Realization

Semantic input

Action = Buy

Agent = Student

Object = Book

Tense = Past

Pseudocode

ALGORITHM SurfaceRealization(SemanticInput)

INPUT: Semantic representation

OUTPUT: Grammatically correct sentence

1. action $\leftarrow$ SemanticInput.Action
2. agent $\leftarrow$ SemanticInput.Agent
3. object $\leftarrow$ SemanticInput.Object
4. tense $\leftarrow$ SemanticInput.Tense
5. verb $\leftarrow$ SelectVerb(action)
6. IF tense = Past:
 verb $\leftarrow$ ConvertToPastTense(verb)
7. subject $\leftarrow$ SelectNoun(agent)
8. noun_phrase $\leftarrow$ SelectNoun(object)
9. sentence_structure $\leftarrow$
 SUBJECT + VERB + OBJECT
10. sentence $\leftarrow$ Realize(
 subject,
 verb,
 noun_phrase
)
11. ApplyCapitalization(sentence)
12. AddFullStop(sentence)

13. RETURN sentence

END ALGORITHM

Processing steps

1. Lexical selection

- Agent = Student $\rightarrow$ **student**
- Action = Buy $\rightarrow$ **buy**
- Object = Book $\rightarrow$ **book**

2. Tense realization

The input specifies past tense:

buy $\rightarrow$ bought

3. Sentence structuring

English follows the basic:

Subject + Verb + Object

Therefore:

Student + bought + book

4. Surface realization

The natural output is:

“The student bought a book.”

Grammatical validation

Feature	Generated output
Subject	The student
Verb	bought
Object	a book
Tense	Past
Word order	Subject–Verb–Object
Agreement	Correct
Punctuation	Correct

The generated sentence is grammatically correct and expresses the complete semantic input.

Q5. Interlingua + Statistical Machine Translation Algorithm

Source sentence:

“The boy is playing football.”

Candidate translations

Language Translation

Tamil சிறுவன் கால்பந்து விளையாடிக் கொண்டிருக்கிறான்.

Telugu బాలుడు ఫుట్బాల్ ఆడుతున్నాడు.

Interlingua representation

EVENT:

ACTION = PLAY

AGENT = BOY

OBJECT = FOOTBALL

TENSE = PRESENT

ASPECT = PROGRESSIVE

Predicate representation:

boy(x) $\wedge$ football(y) $\wedge$

play(x,y) $\wedge$

present(play(x,y)) $\wedge$

progressive(play(x,y))

Candidate generation

Tamil candidates:

1. சிறுவன் கால்பந்து விளையாடுகிறான்.
2. சிறுவன் கால்பந்து விளையாடிக் கொண்டிருக்கிறான்.

Telugu candidates:

1. బాలుడు ఫుట్బాల్ ఆడుతున్నాడు.
2. బాలుడు ఫుట్బాల్ ఆడుతూ ఉన్నాడు.

The system selects the candidate that best preserves the present progressive meaning while maintaining grammatical correctness and natural fluency.

Final translations

Tamil:

“சிறுவன் கால்பந்து விளையாடிக் கொண்டிருக்கிறான்.”

Telugu:

“బాలుడు ఫుట్బాల్ ఆడుతున్నాడు.”

Evaluation

The Interlingua preserves the language-independent meaning:

BOY + PLAY + FOOTBALL + PRESENT PROGRESSIVE

The statistical component then ranks possible Tamil and Telugu realizations using translation probability, grammatical correctness, semantic similarity and target-language fluency.

Thus:

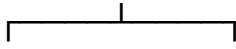
English Source



Semantic Analysis



Language-Independent Interlingua



Tamil



Telugu



Candidate Candidate



Generation Generation



Statistical Statistical



Scoring Scoring



Best Tamil Best Telugu



Translation Translation

This demonstrates that the same Interlingua can generate translations in multiple Indian languages while preserving the original meaning.